Free pythagorean theorem worksheet worksheet 13

Maths | 9th grade | Mixed Operations

1. Solve for x: 3(x - 4) = 2x + 5

A) -1
B) 7
C) 17
D) 4

2. Evaluate: " (81) + 5^2

A) 34
B) 40
C) 44
D) 37

3. If f(x) = 2x^2 - 3x + 1, find f(-2).

A) 15
B) -3
C) 7
D) 3

4. Simplify: (3/4) ÷ (1/2)

A) 1/2
B) 1
C) 3/2
D) 3

5. Solve for y: 4y + 5 = 2(y + 10)

A) 5
B) 15
C) 10
D) -5

6. Find the value of x in the equation: 5x - 3(2x + 9) = 12

A) 15
B) -9
C) 6
D) 3

7. Evaluate the expression: 7 - 2(3 + 4)

A) -7
B) 1
C) 5
D) -1

8. If 3x = 2y and y = 9, find x.

A) 6
B) 5

C) 3
D) 9

9. Simplify: 7(2x - 3) + 5

A) 14x + 26
B) 14x - 16
C) 14x - 21
D) 14x - 10

10. What is the solution of the system: 2a + 3b = 12, 4a - b = 5?

A) (3, 2)
B) (2, 3)
C) (1, 4)
D) (2, 2)

Answer Key

1. B
Distribute on the left: 3x - 12 = 2x + 5. Rearrange to: 3x - 2x = 5 + 12. Simplify to: x = 17.

2. D
" (81) = 9 and 5^2 = 25. So, 9 + 25 = 34.

3. A
f(-2) = 2(-2)^2 - 3(-2) + 1 = 8 + 6 + 1 = 15.

4. C
Dividing by a fraction is the same as multiplying by its reciprocal: (3/4) * (2/1) = 3/2.

5. B
Expand and simplify: 4y + 5 = 2y + 20. Rearrange to: 4y - 2y = 20 - 5. Simplify to: y = 15.

6. C
Expand: 5x - 6x - 27 = 12, simplify to: -x = 39. Solve for x: x = -39.

7. D
Evaluate inside parentheses first: 7 - 2(7) = 7 - 14 = -7.

8. C
Substitute y = 9 into the equation: 3x = 2(9). Solve for x: 3x = 18, x = 6.

9. B
Distribute inside the parentheses: 14x - 21 + 5, simplify to: 14x - 16.

10. A
Solve the system using substitution or elimination method to find a = 3, b = 2.